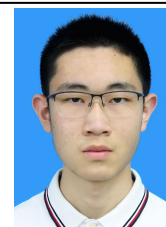


Gao Haiyang

May 22, 2004



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Educational Background

Imperial College London

09/2026-09/2027

Degree: Master of Science (in progress)

Major: Artificial Intelligence

Wuhan University Honor Graduate of Hongyi Honor College

09/2022-06/2026

Degree: Bachelor of Economics & Bachelor of Mathematics

Honours: Honor Graduate of Hongyi Honor College

Major: Finance

Minor: Mathematics and Applied Mathematics

Mark: 3.69/4.0 (Major)

Core Modules: (Numbers in the bracket is course grade out of 100)

Real Analysis (95), Numerical Analysis (95), Stochastic Process (91), Methods of Data Analysis (91), Machine Learning (90), Monetary Banking (96), Corporate Finance (95), Options, Futures and other Derivatives (93), Theory of Markets and Equilibrium (90)

Yale Summer Session

06/2025-08/2025

Courses: Game Theory (course grade: A), YData: Introduction to Data Science (course grade: A)

Graduate Record Examination

General Test Scores: 322

Published Papers

2024 Green Finance Annual Report, 2025, Second author, Industry Publication

Papers in Preparation

Sparseformer-Based Electric Load Time-Series Forecasting

07/2025-03/2026

Position: Researcher, supervised by Dr. Shivam Sharma, Yale University

- Led the design and manuscript drafting of a Sparseformer model for electric-load time-series forecasting, addressing vanilla Transformer's $O(L^2)$ self-attention cost, memory bottlenecks, slow long-horizon inference, and limited effective receptive fields through Transformable Sparse Attention with offset-network sampling and STL seasonal-trend decomposition features.
- Built the PyTorch training and evaluation pipeline on ETTh1, WTH, and ECL, benchmarked against Informer, Reformer, and DeepAR with MSE/MAE, and achieved consistently higher forecasting accuracy; manuscript available [Here](#)

Academic Projects

Economic Effects of LA 2028 Olympics

08/2025-03/2026

Position: Research Assistant, supervised by Dr. Lorena Martin, Trojans Sports Research Lab, USC

- Participating in a research project on analysing the effect of LA 2028 Olympics to local economy.

Innovation and Pricing Research on Corporate Carbon Financial Products

03/2025-07/2025

Position: Research Assistant

Sponsor: Research Group of Professor Ren Yujie, International Institute of Green Finance (IIGF), CUFE

- Participated in project Innovation and Pricing Research of Corporate Carbon Financial Products sponsored by IIGF, CUFE and State Grid Research Institute; attended weekly meetings and contributed quantitative finance expertise to address pricing challenges.

Research on Platform Economy Theory

09/2024 - 07/2025

Position: Research Assistant

Sponsor: Wuhan University, Research Group of Professor Shen Bo

- Responsible for daily screening of current situation of platform economy, collected information on development trends.
- Participated in exploring platform business models, market competition and user behaviour; discussed impact of platform economy on economic development.

Teaching Assistant for Game Theory Module in Wuhan University**02/2025-07/2025**

Position: Teaching Assistant

- Assisted instructor in Game Theory course, collected and graded assignments monthly, selected as an outstanding TA, earning high recognition from both the instructor and students.
- Provided in-depth explanations during exercise classes, including addressing students' questions on Cournot equilibrium and other core models to strengthen their understanding of theoretical concepts.
- Organised Q & A sessions before quizzes and exams; supervised and graded mid-term and final exams, as well as collecting and grading monthly assignments.

Industry Experience (Internships)**Industrial and Commercial Bank of China, Paid, Full-Time****08/2025-10/2025**

Position: Intern, Corporate Finance Department & International Business Department

- Served at the corporate finance department as an intern to learn the loan approval and customer credit evaluation process, analysed the existing market and financial risks.
- Analysed various financial instruments and documents used in trade financing services, such as letters of credit, import and export bills of lading, bills of exchange, and documentary collections, as well as evaluated their roles in mitigating risk and facilitating international trade transactions.

China Securities Co., Ltd, Full-Time**07/2024-08/2024**

Position: Intern, Bond Underwriting Department

- Automated the processing of Wind database using Python, enabling the completion of weekly bond reports 40% with higher precision; published 5 weekly reports to provided interested parties with valid information for strategy adjustment.
- Supported statistical analysis for partnerships; completed automated data processing via Python and assisted in bond market data analysis with Wind to provide data support to team and help optimise business cooperation strategies.

KPMG Huazhen LLP, Paid, Full-Time**01/2024-02/2024**

Position: Intern, Audit Department

- Participated in offline annual review of CRRC Corporation, experiencing entire audit process from determining audit objectives to composing audit report; developed Excel templates to automate data import and validation, improved team efficiency by 70%.
- Established communication channels between audit project team and client companies through inspecting audit objectives and collecting information; understood audit risks identification methods and audit risk models.

Languages

Mandarin Chinese (native), English (proficient)

IELTS: Overall: 7.5 (Listening: 7.5, Reading: 9.0, Writing: 6.5, Speaking: 7.5)

Programming Skills

Python, The R Programming Language, STATA, LaTeX, MATLAB, Excel

Project Experience**Self-developed WeChat Mini Program**

A badminton-focused mini program built independently for match recording, Elo ratings, ranking comparison, and performance trend tracking. Currently have more than 500 users.

**Honours & Awards**

- Outstanding Student of Hongyi Honor College for academic year 2024-2025
- 2nd class scholarship for academic year 2024-2025
- 3rd class scholarship for academic year 2023-2024
- 3rd Prize in National ETF Simulation Investment Elite Challenge
- Outstanding Graduation Thesis for academic year 2025-2026
- Won team championship 2 times as a key player, 2nd Place in Men's Doubles and 2nd Place in Mixed Doubles, Hongyi Honor College Badminton Tournament
- 3rd Place in university-level football tournament, MVP; 1st Place in college-level football tournament, MVP and Golden Glove Award